Multimodal Language Models

Sepideh Mamooler, 20.05.2026

- Ross says, “We were on a break”! Based on Rachel's face, does she agree?
- What time of day is this scene set?
- Analyse the acting in this video.

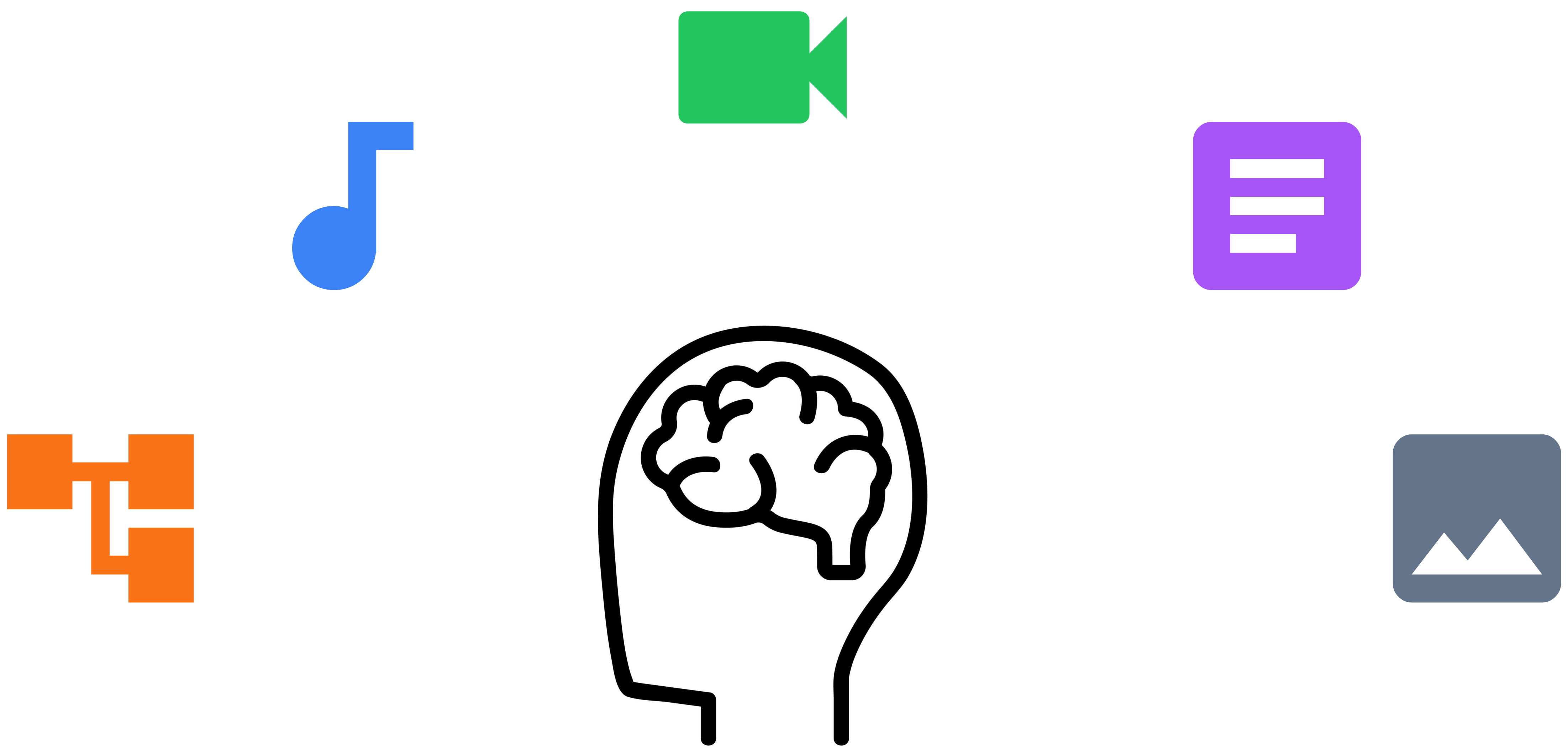
video frames
audio
text

Qwen3-Omni

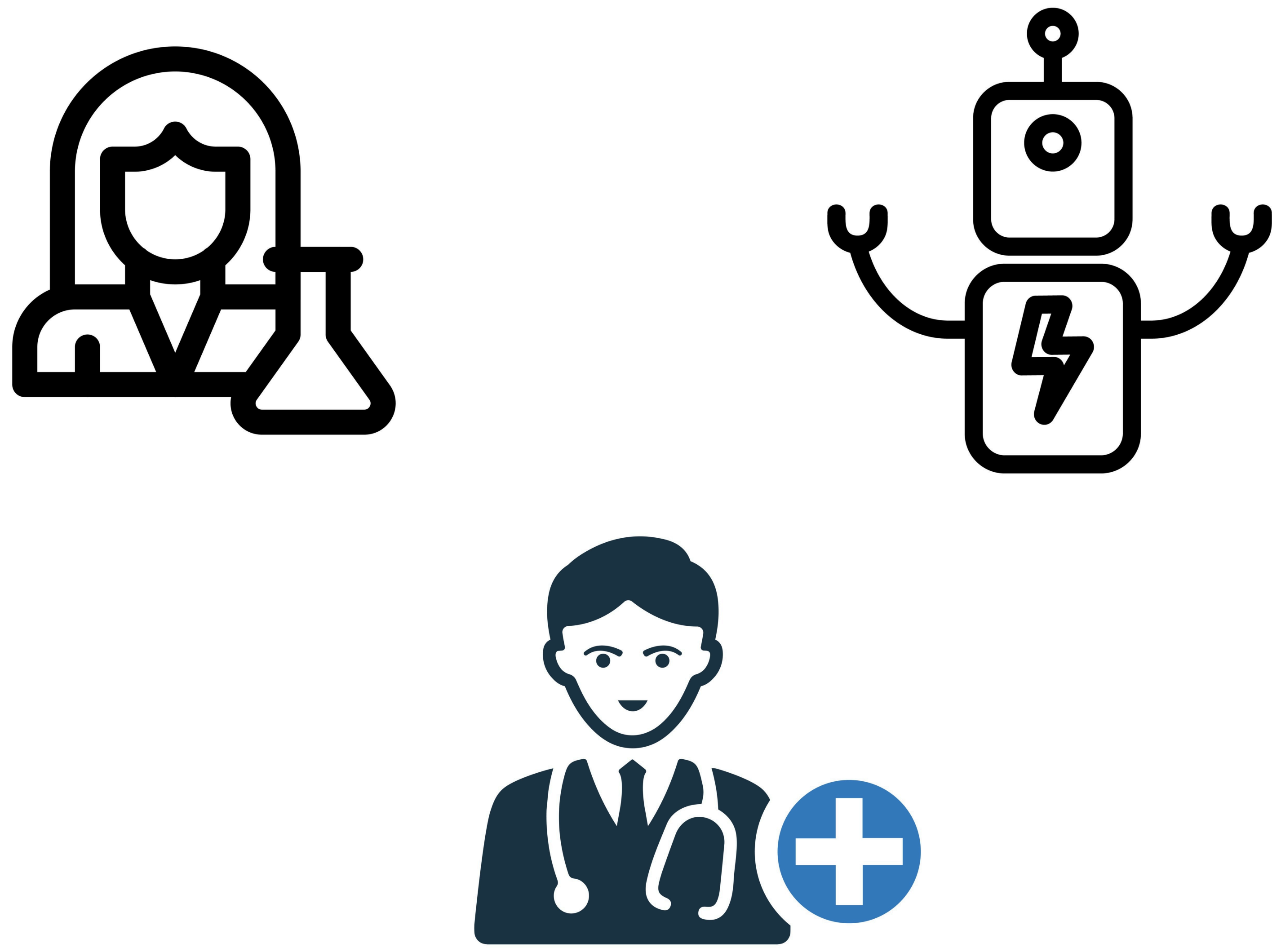
Rachel’s face says it all—she doesn’t agree at all. Her expression is tight with frustration, her eyebrows are pulled together, and her mouth is set in a way that suggests she’s done listening to that excuse. There’s no softness or understanding in her eyes, just clear annoyance and a hint of hurt. It’s the kind of look you get when you’ve heard the same thing too many times and you’re finally done pretending it makes sense. She’s not just disagreeing—she’s shutting it down.

Why Multimodality?

Human perception is inherently multimodal.



Many real tasks are multimodal by nature.



What Counts As Modality?

From An Engineer's Perspective

Text — Discrete and sequential

Image — Continuous and 2D-spatial

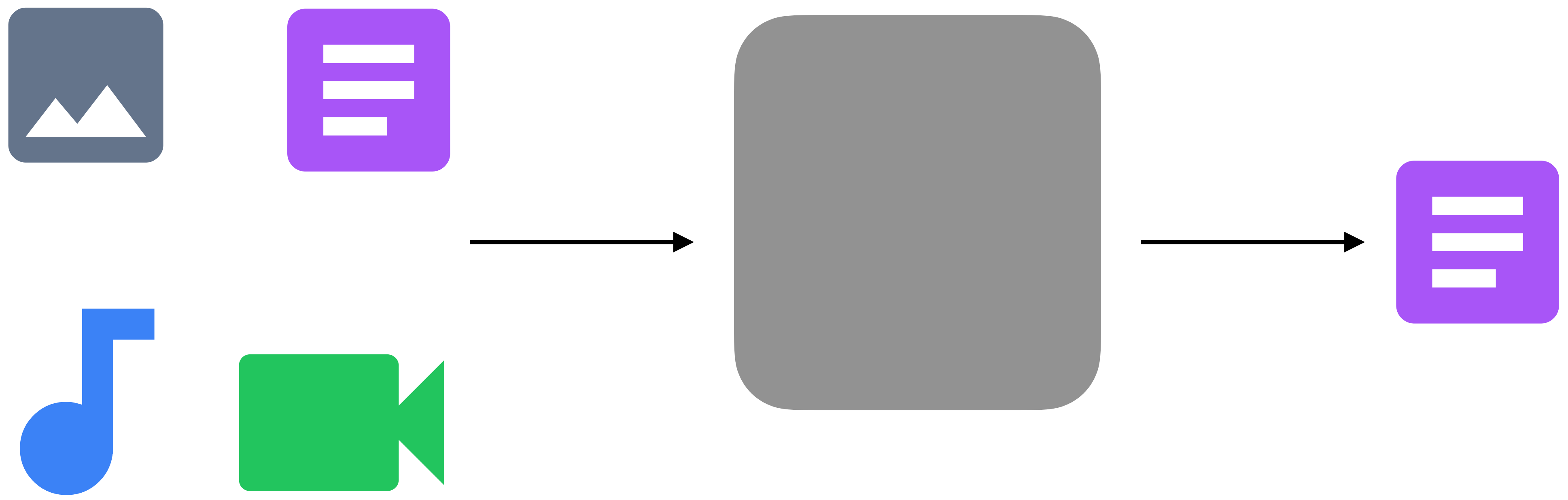
Audio — Continuous and 1D-temporal

Video — Image and temporal axis + Audio

A type of information with a **distinct structure** that **requires its own encoding strategy**.

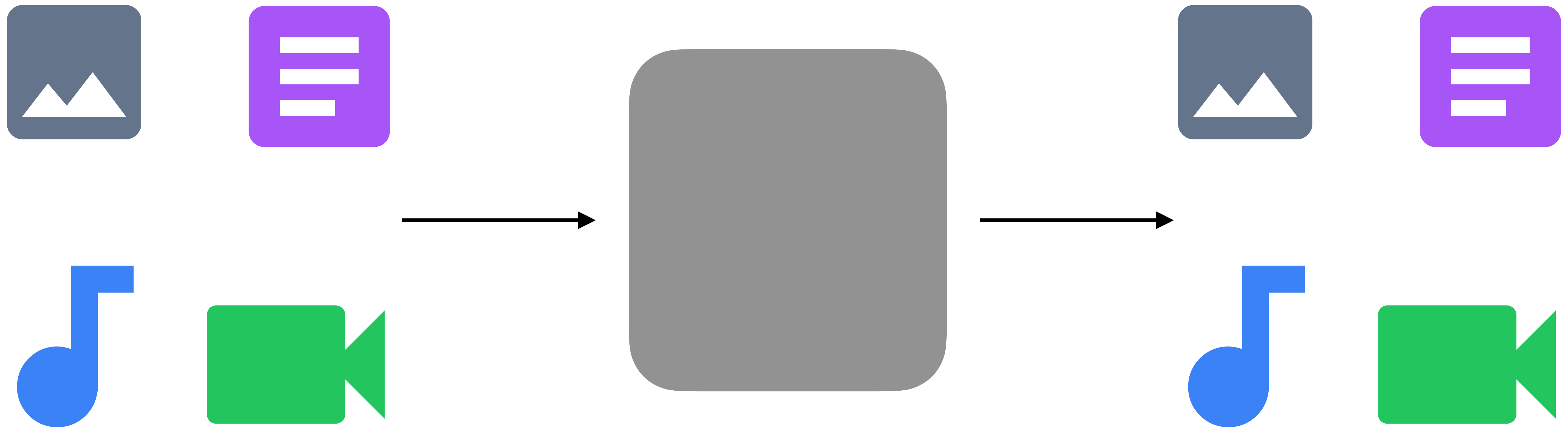
Flavours of Multimodality

Understanding:
Any modality -> Text

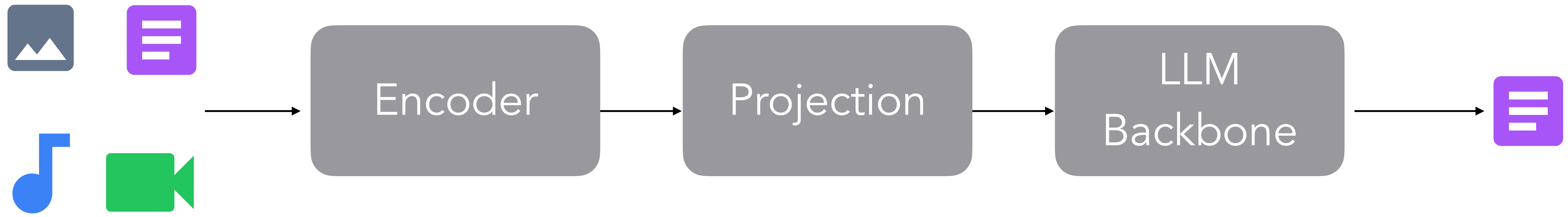


Main focus for today

Generation:
Any modality -> Any modality

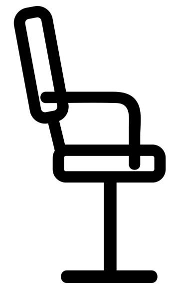


The Big Picture



Key Challenges

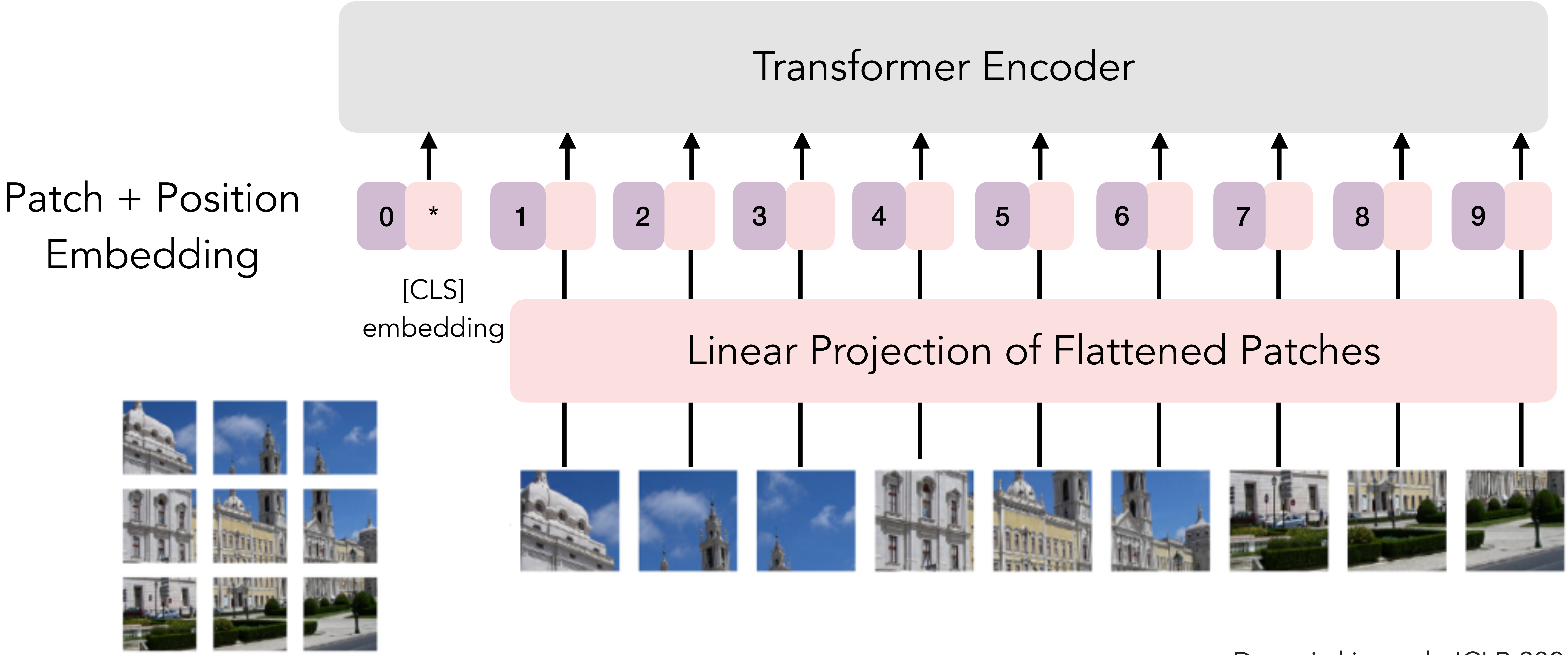
Representation: How to encode different modalities?

Alignment: How to make a model know “chair” and  refer to the same thing?

Bolt-on Training: How to turn LLMs into multimodal LLMs?

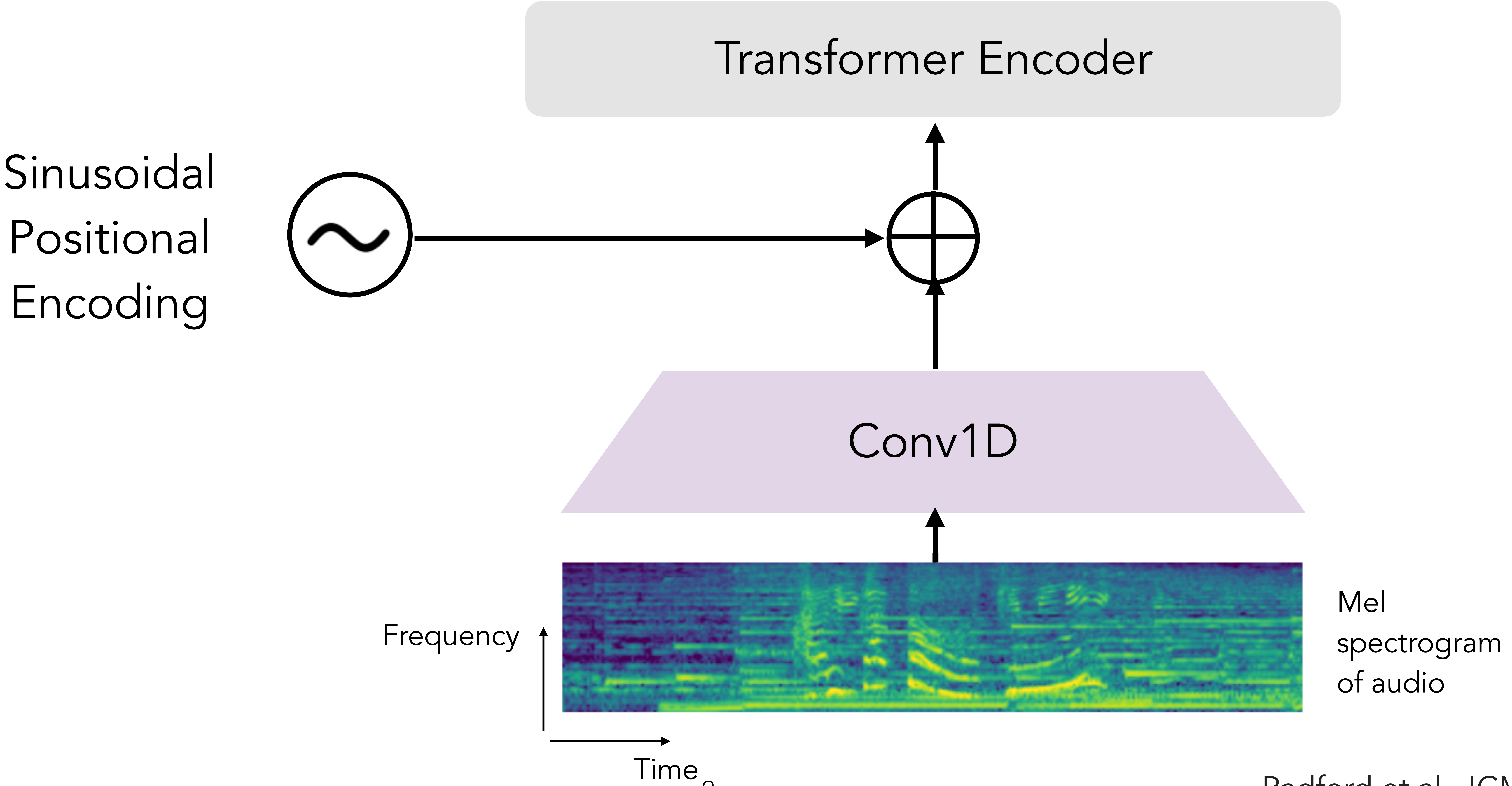
Native Training: How to train models that are multimodal from the start?

Encoder — Image



Dosovitskiy et al., ICLR 2021

Encoder — Audio

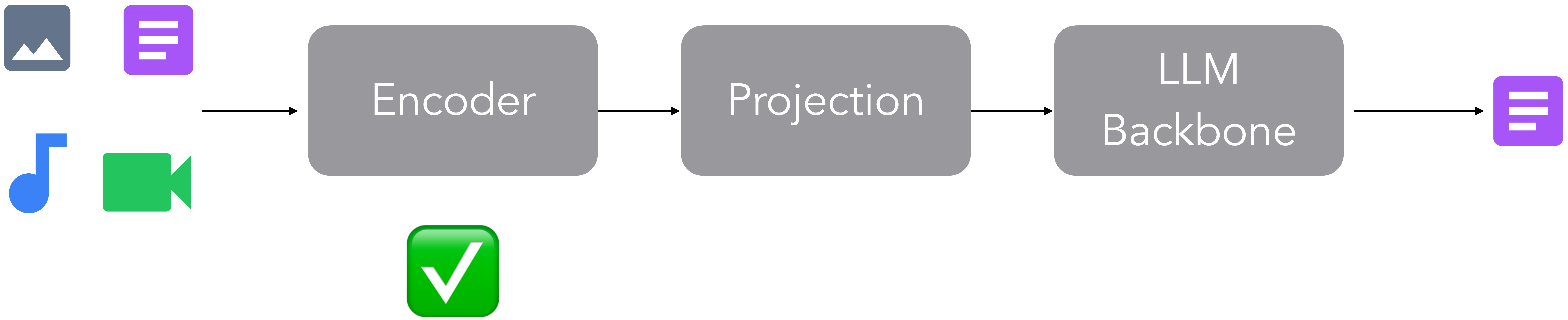


Radford et al., ICML 2023

Encoder — Video

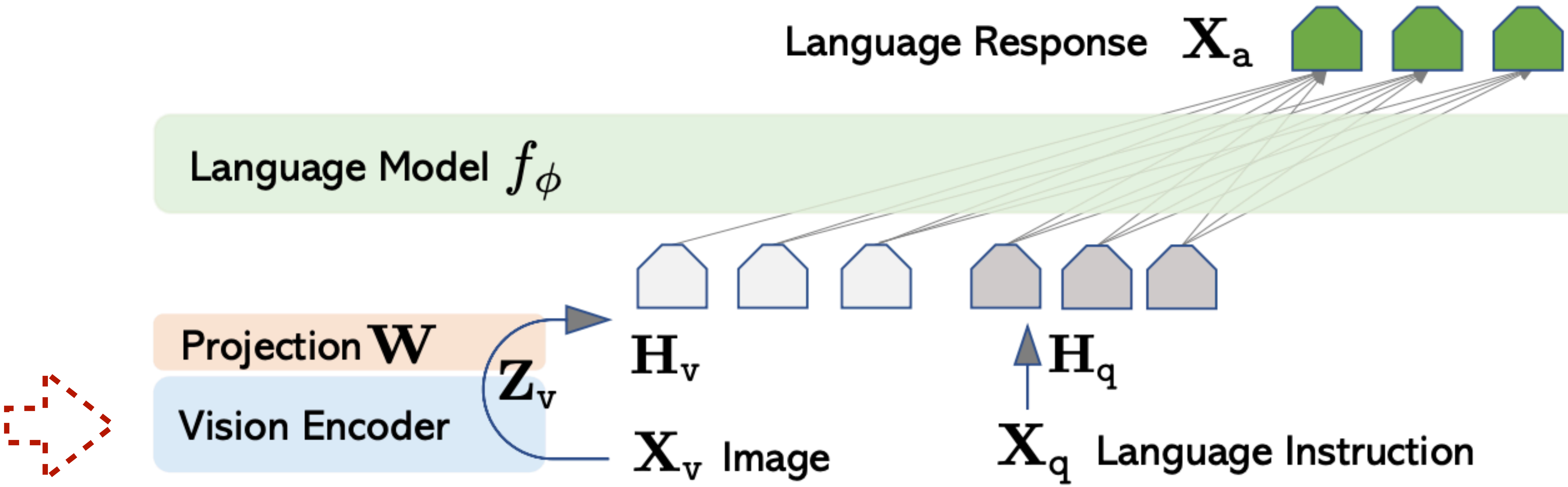
Frame Selection

The Big Picture



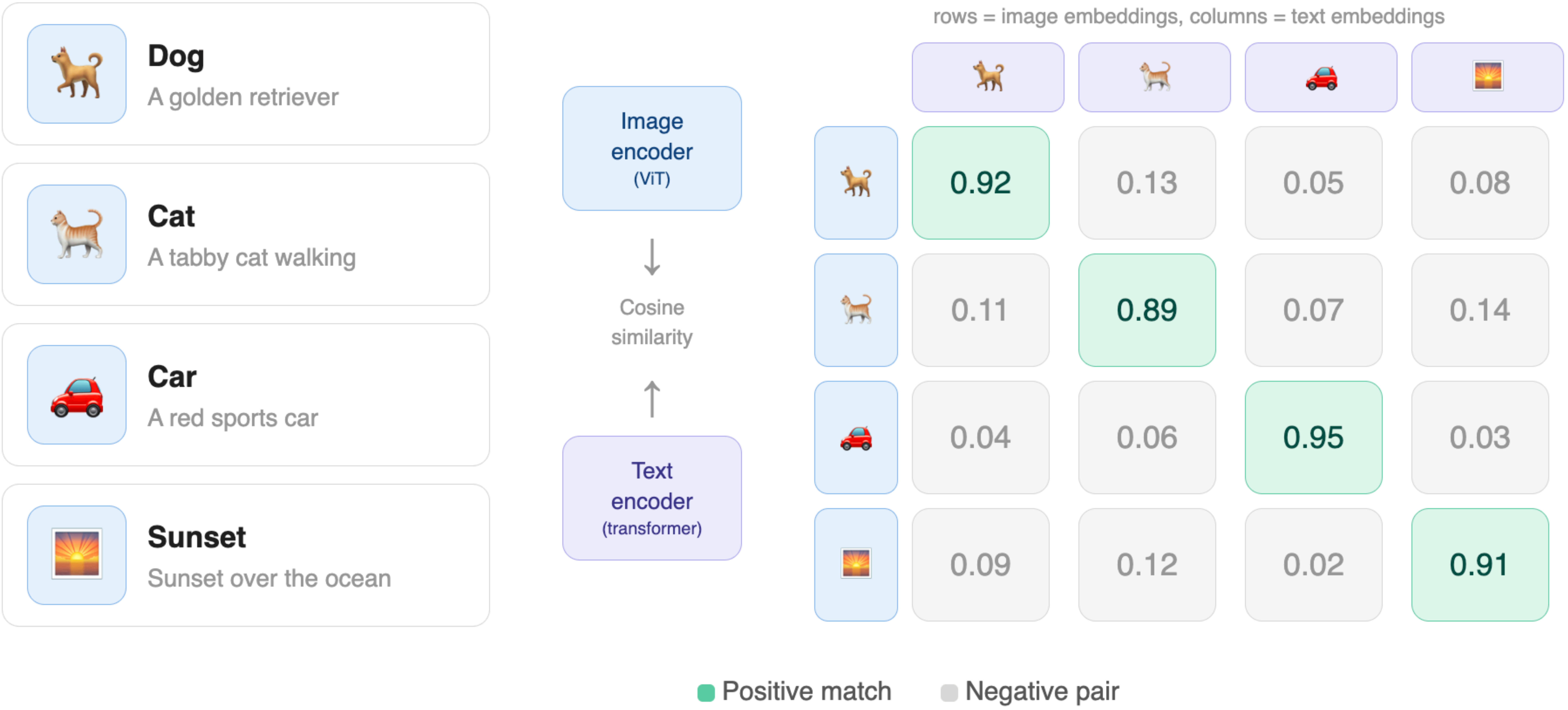
LLaVA

Large Language and Vision Assistant



Liu et al., NeurIPS 2023

Vision Encoder - CLIP



Radford et al., ICML 2021

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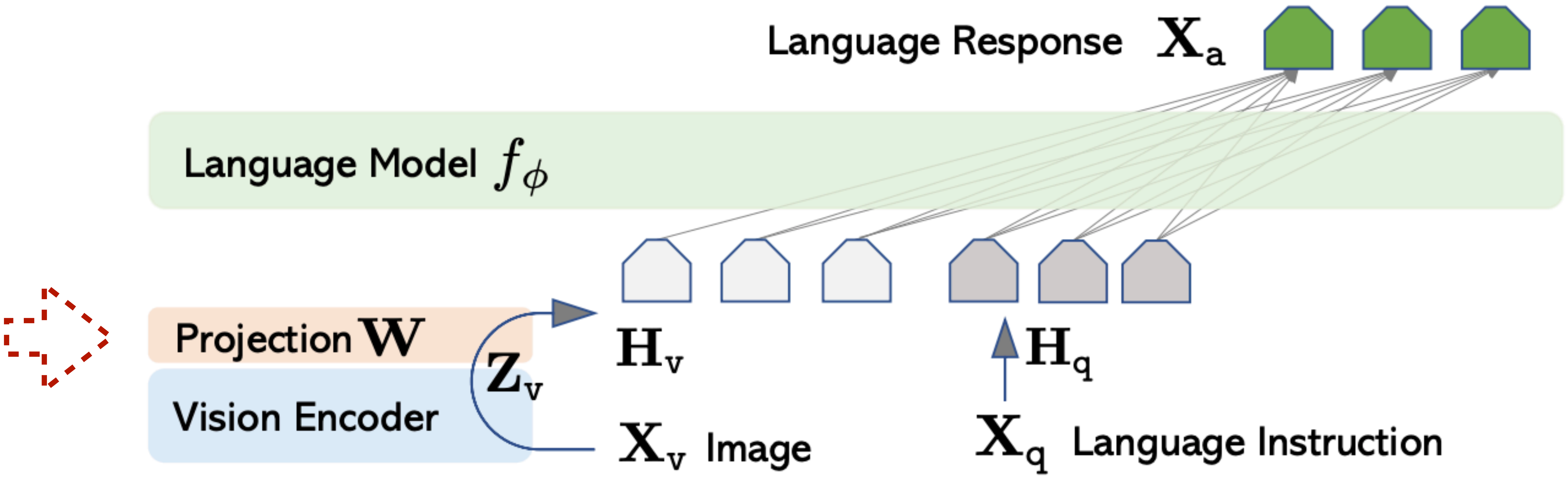
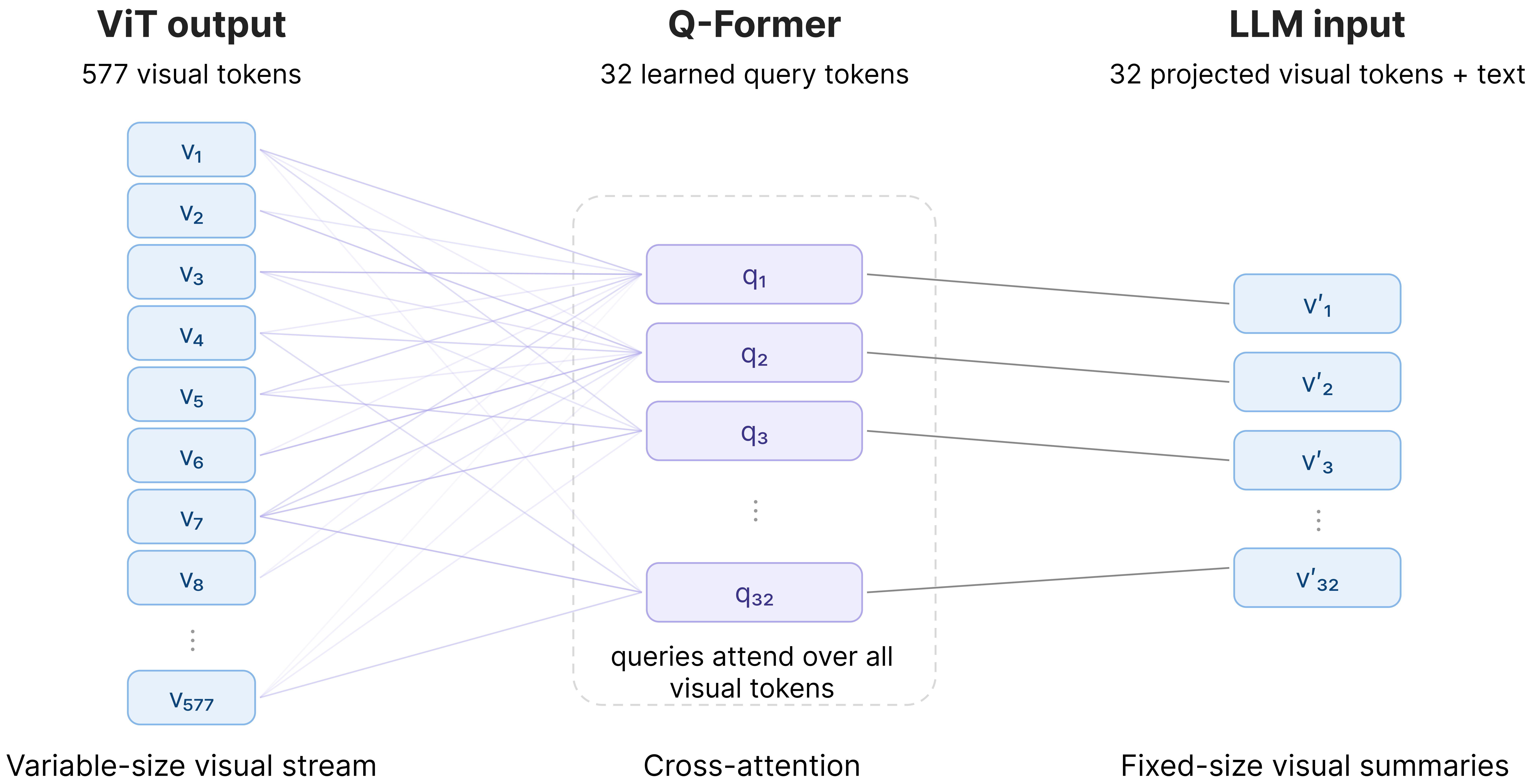


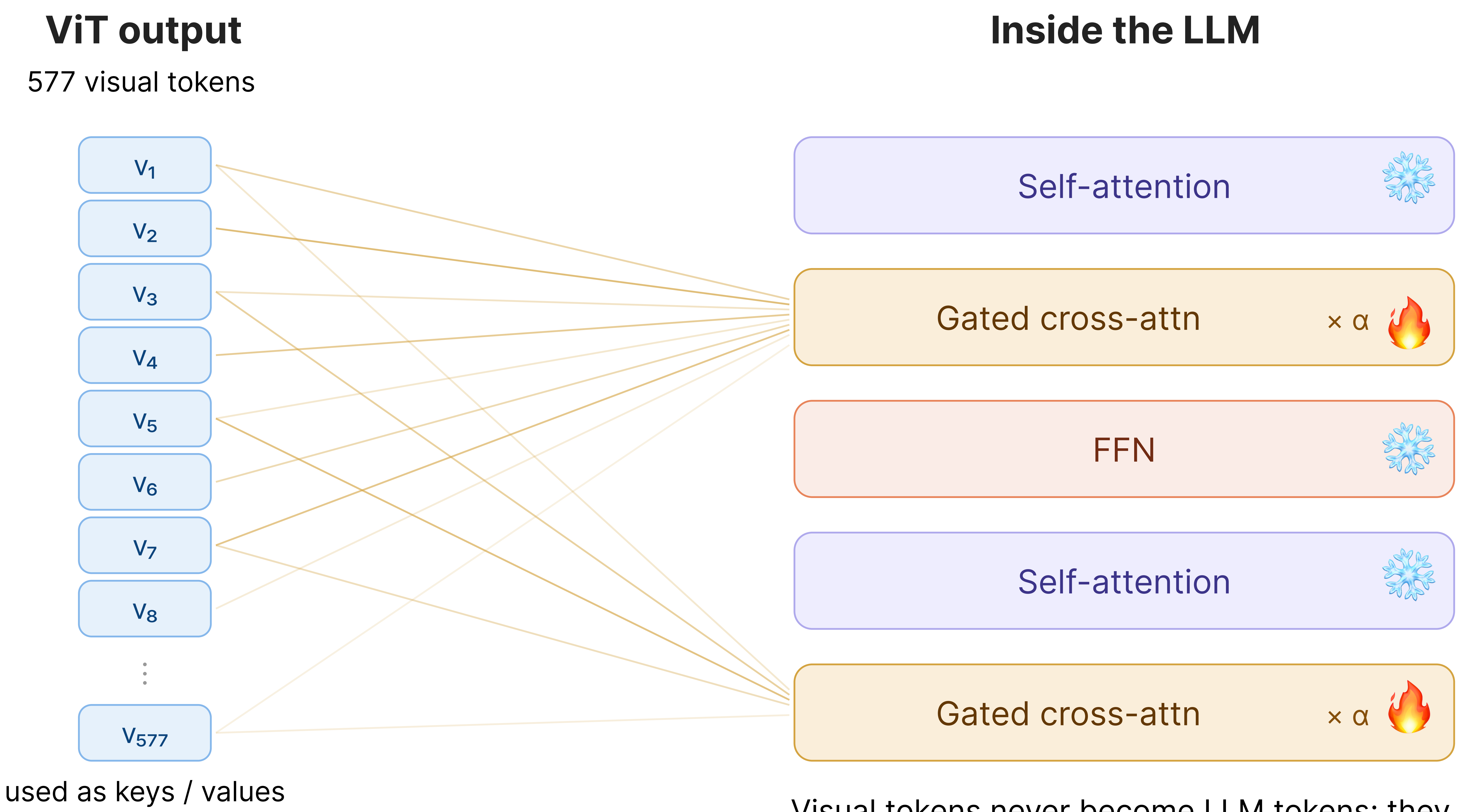
Liu et al., NeurIPS 2023

Projection Layer - Q-Former



Li et al., ICML 2023

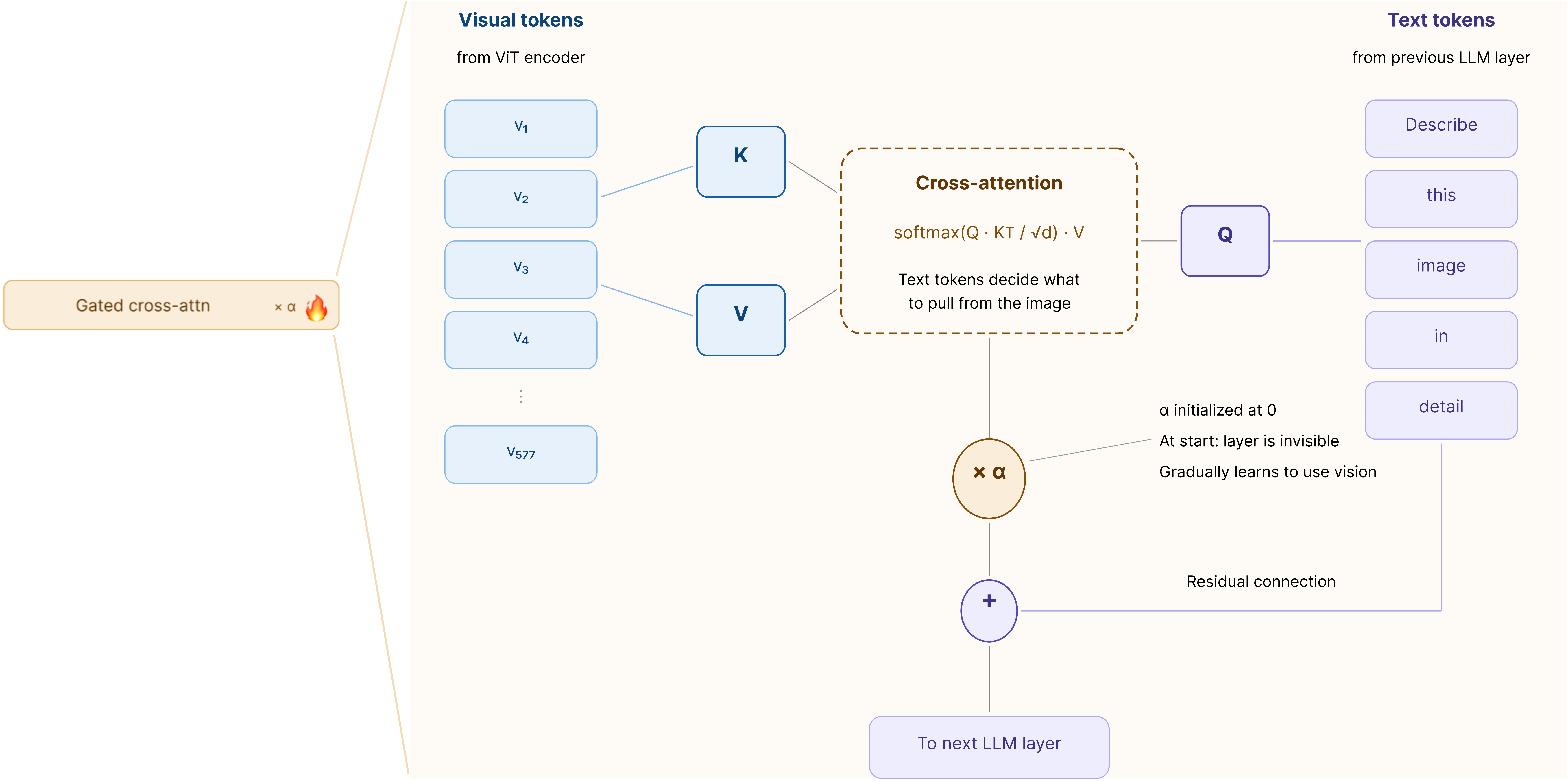
Projection Layer - Gated Cross Attention



Visual tokens never become LLM tokens; they stay external and are queried on demand by the text tokens.

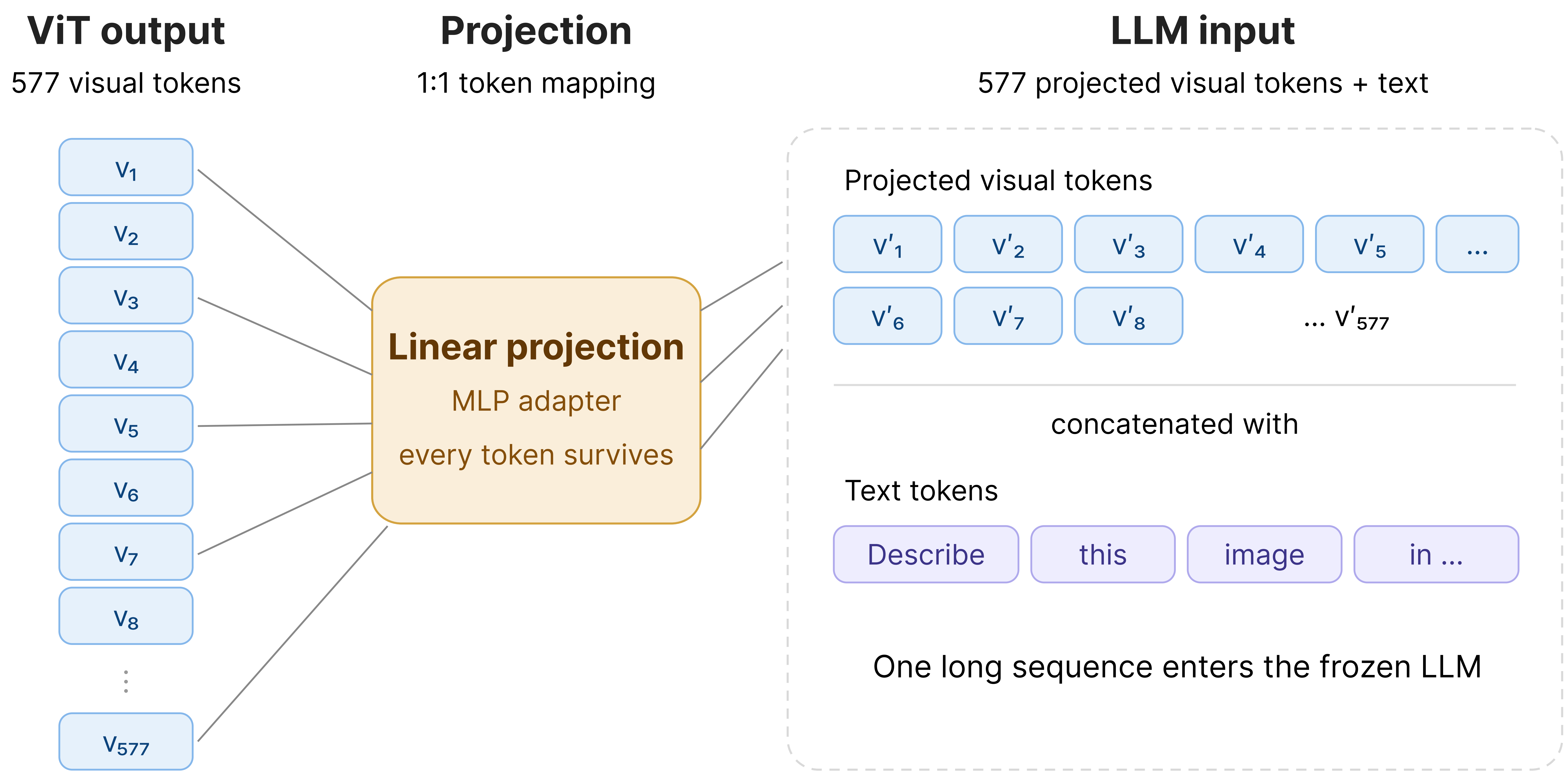
Alayrac et al., NeurIPS 2022

Projection Layer - Gated Cross Attention



Alayrac et al., NeurIPS 2022

Projection Layer - Linear

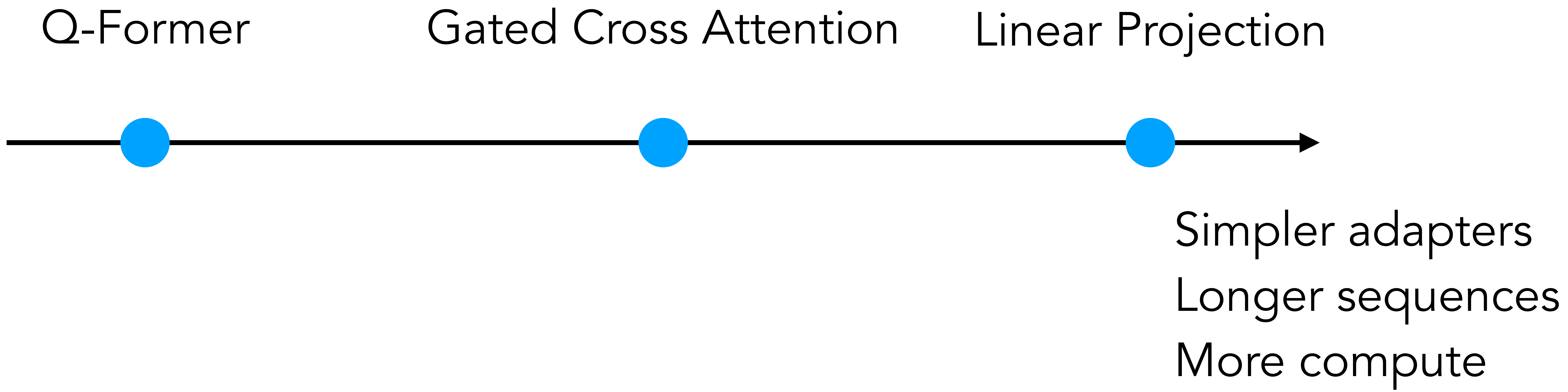


Projected visual tokens stay one-for-one with the original patches

Liu et al., NeurIPS 2023

Projection Layer

The less you compress the vision before the LLM, the more you trust the LLM to figure it out!



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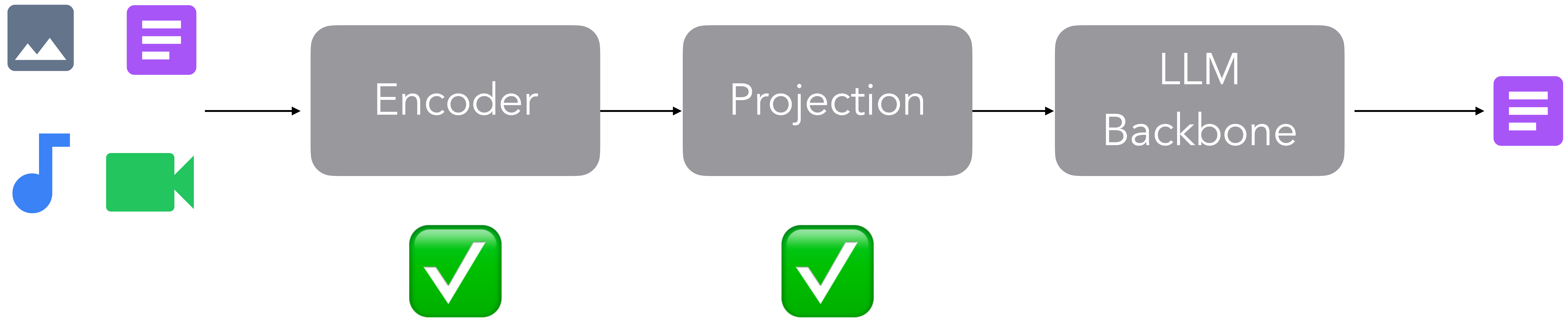

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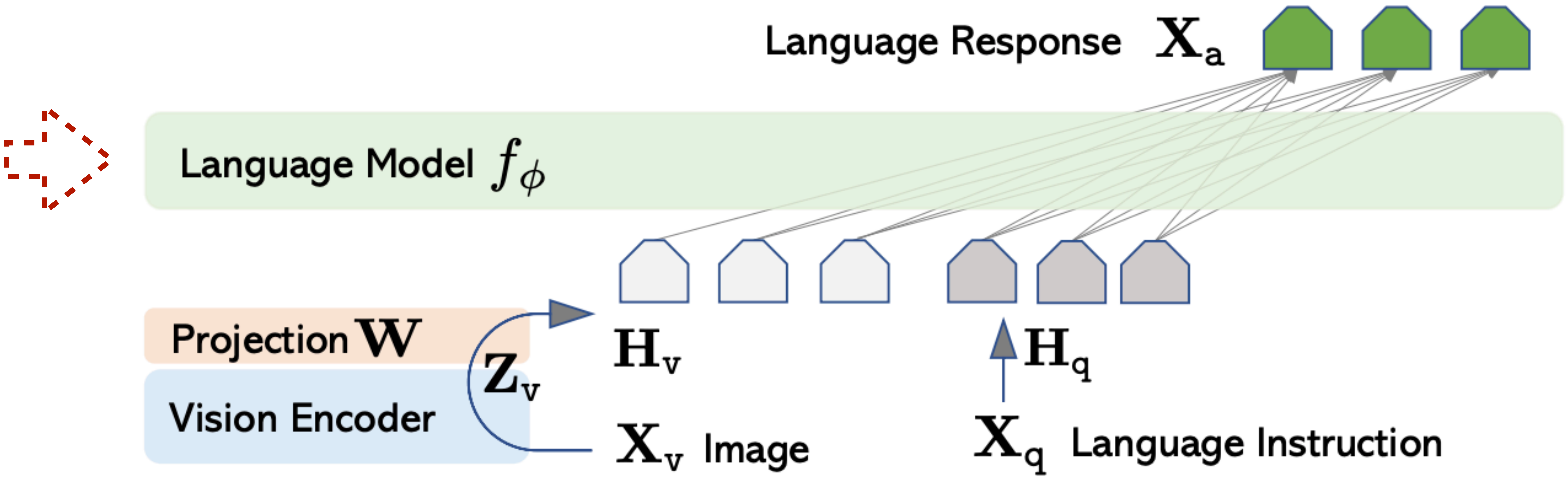


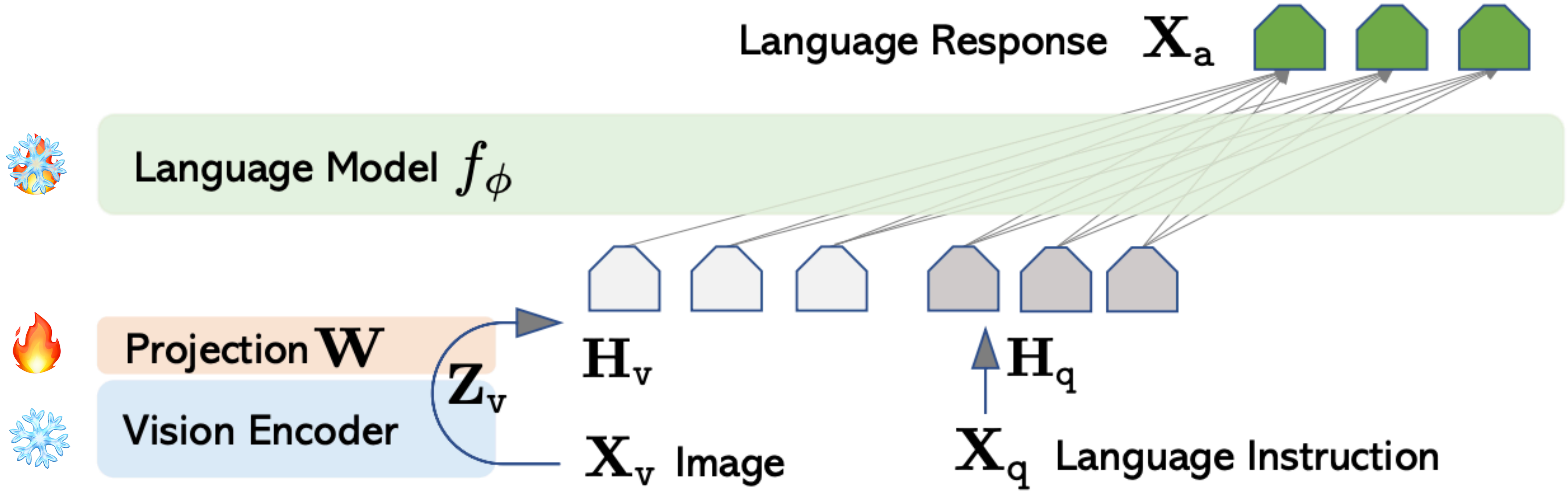
Liu et al., NeurIPS 2023

LLaVA - Training Recipe

Align First, Then Instruction-tune

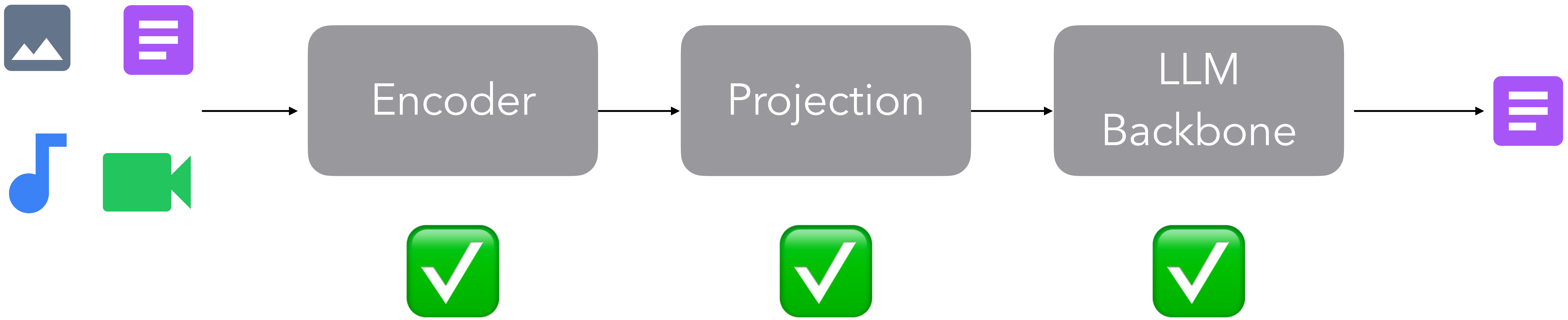
Stage 1: Align
Training data: image-caption pairs

Stage 2: Instruction-tune
Training data: visual QA, conversation about images, etc



Liu et al., NeurIPS 2023

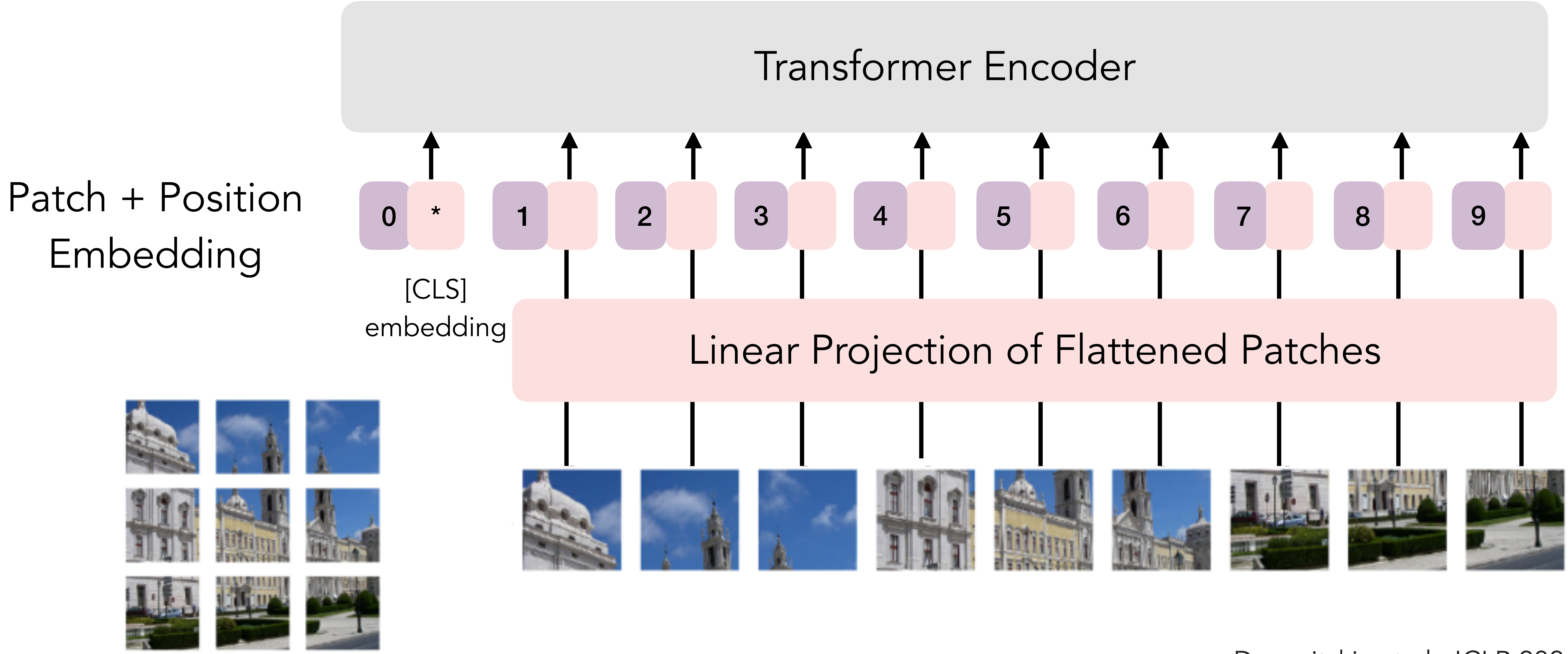
The Big Picture



This is the blueprint behind most multimodal models today:

ViT + MLP + LLM
with multi-stage training

Encoder — Image



Dosovitskiy et al., ICLR 2021

Not Every Image Deserves 577 Tokens!

Resize to 336 x 336

Resize to 336 x 336

One token for just one color!

14 x 14 patches each patch = one token

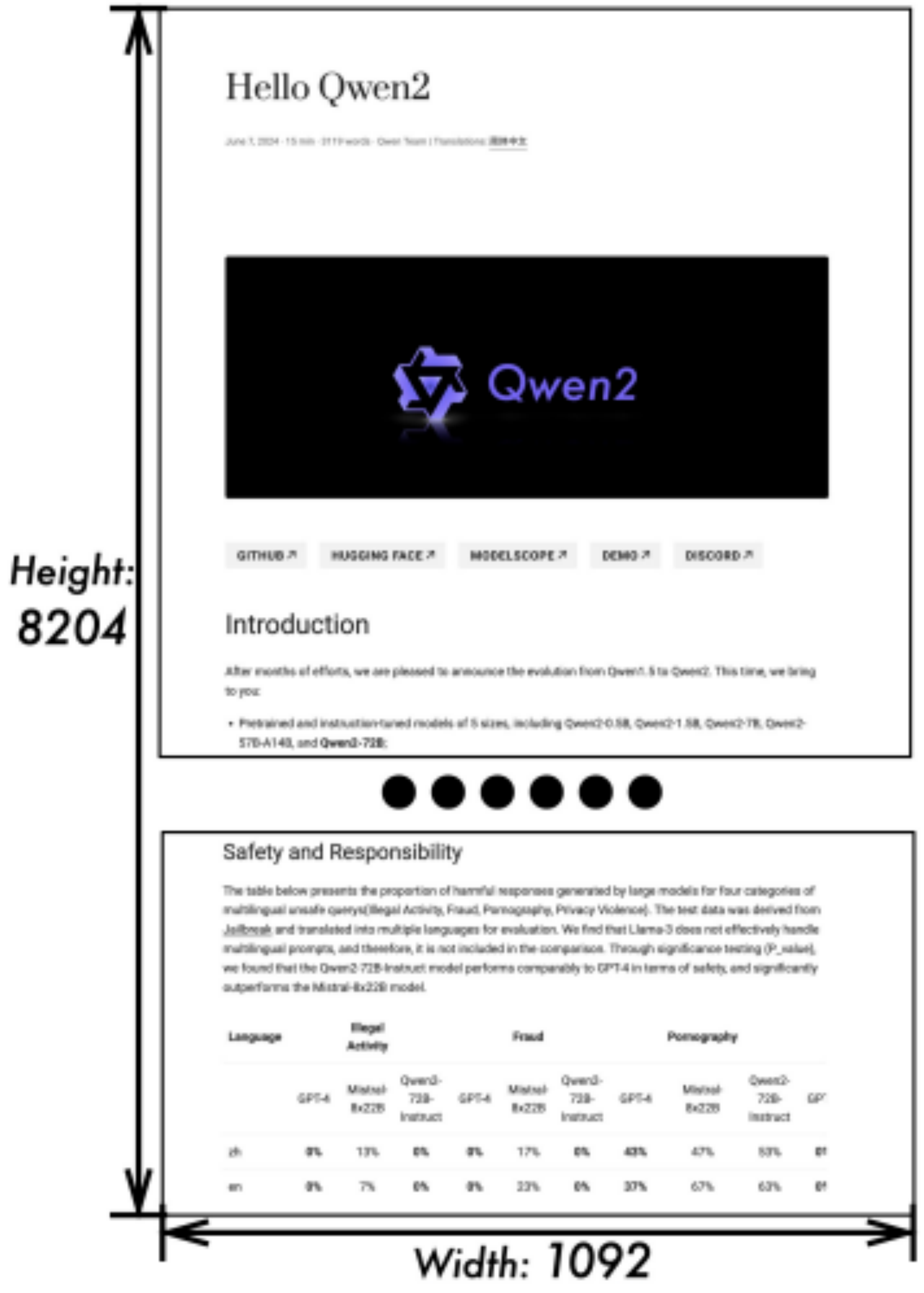
All of this information encoded in only one token!

Dynamic Resolution

QwenVL's solution:

Use the image's native resolution

Higher resolution => more tokens

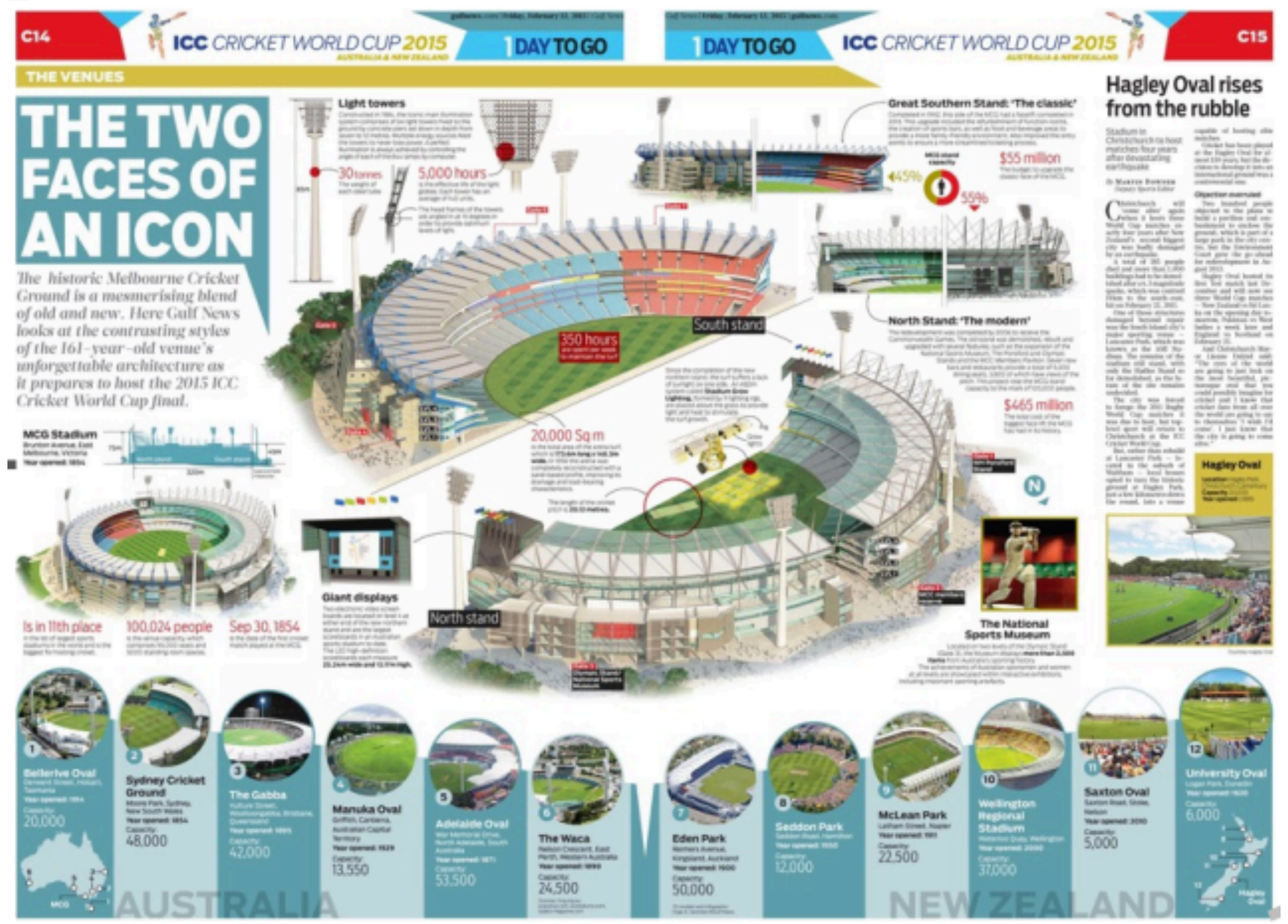


→ 11427 tokens

→ 1125 tokens

Qwen2-VL, 2024

Dynamic Resolution



InternVL's solution:
Split the image into tiles of fixed size, encode each tile individually with 576 tokens

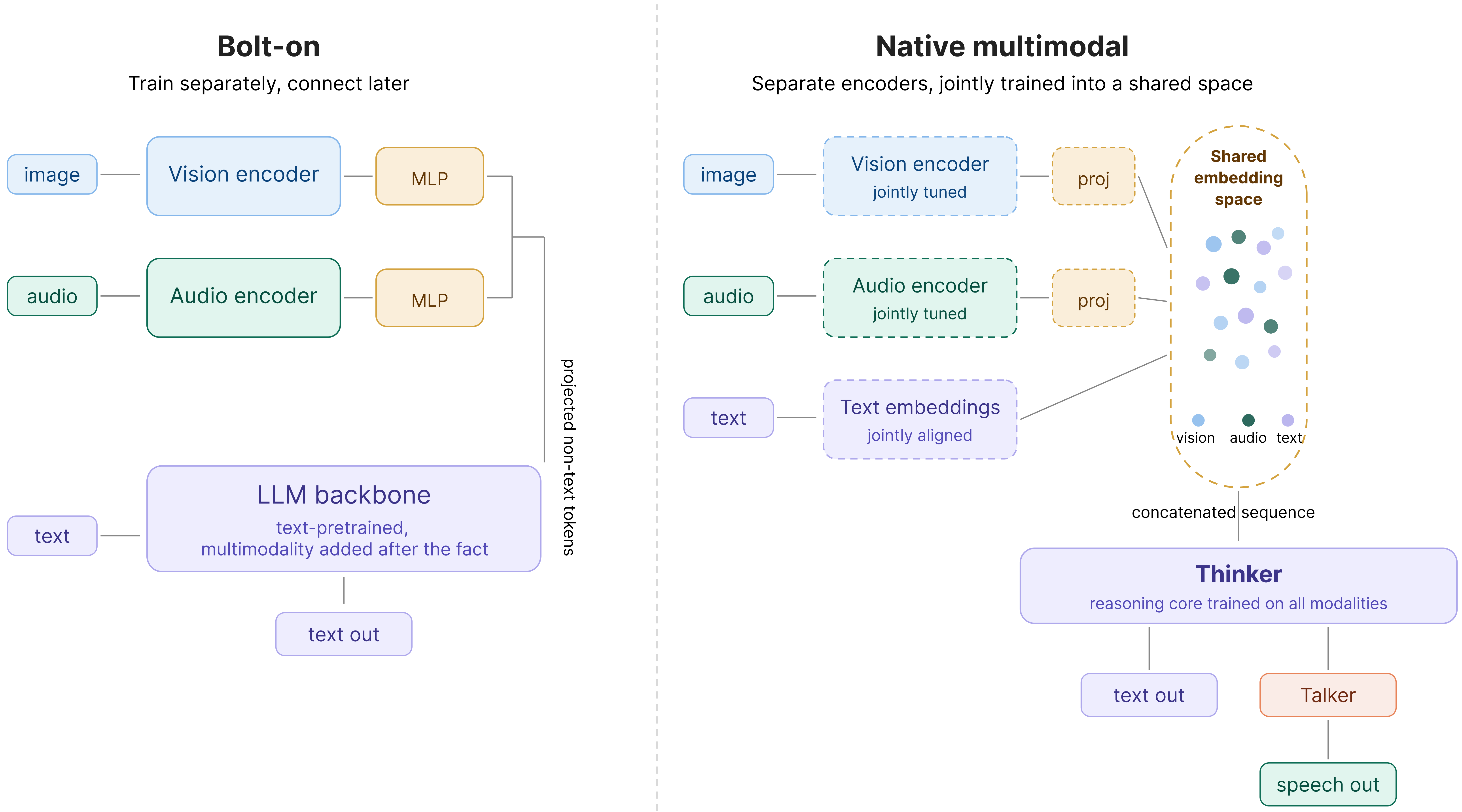
Total num_tokens = num_tiles x 576

InternVL2.5, 2025

What If the Model Was Multimodal From The Start?

Training recipe

1. Train encoders separately
2. Train adapter
3. Fine-tune LLM+adapter



Training recipe

1. Start with text
2. Gradually mix in all modalities

Qwen3-Omni, 2025

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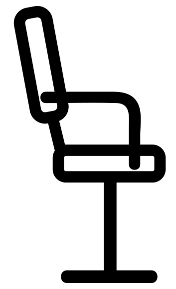
Alignment: How to make a model know "chair" and  refer to the same thing?

Bolt-on Training: How to turn LLMs into multimodal LLMs?

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Evaluation

How Do We Know It Works?

What's tested	Example benchmarks	Format
Vision Reasoning	MMMU, DocVQA, ChartQA	MCQ / free-form
Audio Reasoning	MMAU, VoiceBench	MCQ / open-ended
Audiovisual Reasoning	WorldSense, OmniBench, DailyOmni	MCQ
Text Performance	MMLU, GPQA, AIME	MCQ / numerical

Open Problems

Hallucination: Is this a failure of the vision encoder, the projection, or the LLM?

Visual Token efficiency: Humans don't process every pixel equally; we attend selectively. Can models learn to do the same?

Native vs bolt-on training: Bolt-on models with good adapters sometimes match native models on benchmarks, at a fraction of the training cost. And native models still start with text-first pretraining anyway, so how "native" are they really?

And many others!

Reading List

Tokenization: How Modalities Become Sequences

An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale Dosovitskiy et al., 2021 The Vision Transformer (ViT) paper. Introduces patch-based tokenization of images and applies the transformer architecture to vision. The foundation for every visual encoder discussed in this lecture. https://arxiv.org/abs/2010.11929

Robust Speech Recognition via Large-Scale Weak Supervision Radford et al., 2022 The Whisper paper. Describes the audio encoder architecture (mel spectrogram → convolutional stem → transformer) that became the standard for audio processing in multimodal models. https://arxiv.org/abs/2212.04356

Reading List

Contrastive Alignment: CLIP & SigLIP

Learning Transferable Visual Models From Natural Language Supervision Radford et al., 2021 The CLIP paper. Introduces contrastive pretraining on image-text pairs to learn a shared embedding space. The vision encoder from CLIP is used directly in LLaVA and many other multimodal LLMs. https://arxiv.org/abs/2103.00020

Sigmoid Loss for Language Image Pre-Training Zhai et al., 2023 The SigLIP paper. Replaces CLIP's softmax-based contrastive loss with a per-pair sigmoid loss, enabling better scaling and stronger performance at smaller batch sizes. https://arxiv.org/abs/2303.15343

Reading List

<u>**Connecting Vision to LLMs: Adapter Architectures**</u>

Flamingo: A Visual Language Model for Few-Shot Learning Alayrac et al., 2022 (DeepMind) Introduces gated cross-attention layers inserted between frozen LLM layers, allowing the LLM to attend to visual features. The conservative approach: modify the LLM architecture minimally. https://arxiv.org/abs/2204.14198

BLIP-2: Bootstrapping Language-Image Pre-training with Frozen Image Encoders and Large Language Models Li et al., 2023 (Salesforce) Introduces the Q-Former, a lightweight querying transformer that compresses visual tokens into a fixed-size representation. The bottleneck approach: compress first, then feed to the LLM. https://arxiv.org/abs/2301.12597

Visual Instruction Tuning Liu et al., 2023 The LLaVA paper. Demonstrates that a simple linear projection from CLIP visual features to the LLM embedding space, combined with instruction tuning, achieves strong multimodal performance. The simple approach that works. https://arxiv.org/abs/2304.08485

Improved Baselines with Visual Instruction Tuning Liu et al., 2023 LLaVA-1.5. Upgrades the linear projection to a two-layer MLP and introduces academic-task-oriented data, establishing the architecture that most subsequent work builds on. https://arxiv.org/abs/2310.03744

Reading List

<u>**Scaling: Dynamic Resolution and Beyond**</u>

Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution Wang et al., 2024 (Alibaba) Introduces native dynamic resolution processing with 2D-RoPE position embeddings, allowing the ViT to handle variable-size inputs without fixed tiling. https://arxiv.org/abs/2409.12191

InternVL: Scaling up Vision Foundation Models and Aligning for Generic Visual-Linguistic Tasks Chen et al., 2023 (Shanghai AI Lab) Introduces InternViT-6B and the tile-based dynamic resolution approach: fixed tile size (448×448), variable number of tiles based on image resolution. https://arxiv.org/abs/2312.14238

How Far Are We to GPT-4V? Closing the Gap to Commercial Multimodal Models with Open-Source Suites Chen et al., 2024 (Shanghai AI Lab) The InternVL 1.5/2 paper. Scales up the dynamic tiling approach and demonstrates competitive performance with commercial models. https://arxiv.org/abs/2404.16821

Reading List

Native Multimodal Pretraining and Evaluation

Qwen3-Omni Technical Report Qwen Team, 2025 (Alibaba) Introduces the Thinker-Talker architecture with MoE for native multimodal pretraining. Demonstrates that natively training on all modalities can match single-modality performance without regression. https://arxiv.org/abs/2509.17765